

TrES-3b

R-0377

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_250618 · created 2026-07-18T18:49:03+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460844.8489 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.021
Transit depth (Rp/Rs)^2	1.97 +/- 0.58 [%]
Orbital Inclination (inc)	86.97 +/- 1.66
Airmass coefficient 1 (a1)	1.126 +/- 0.017
Airmass coefficient 2 (a2)	-0.106 +/- 0.049
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	#3 - [294, 329]
Optimal Method	PSF photometry
Transit Duration (day)	0.0784 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.14 ± 0.021 vs 0.16309 (deviation 1.1σ)
Duration fit vs published	0.0784 d vs 0.0567 d (deviation 4.94σ)
Mid-time O–C	-0.7 min
Depth SNR	6.7
Residual scatter	1.51 %

Light curve

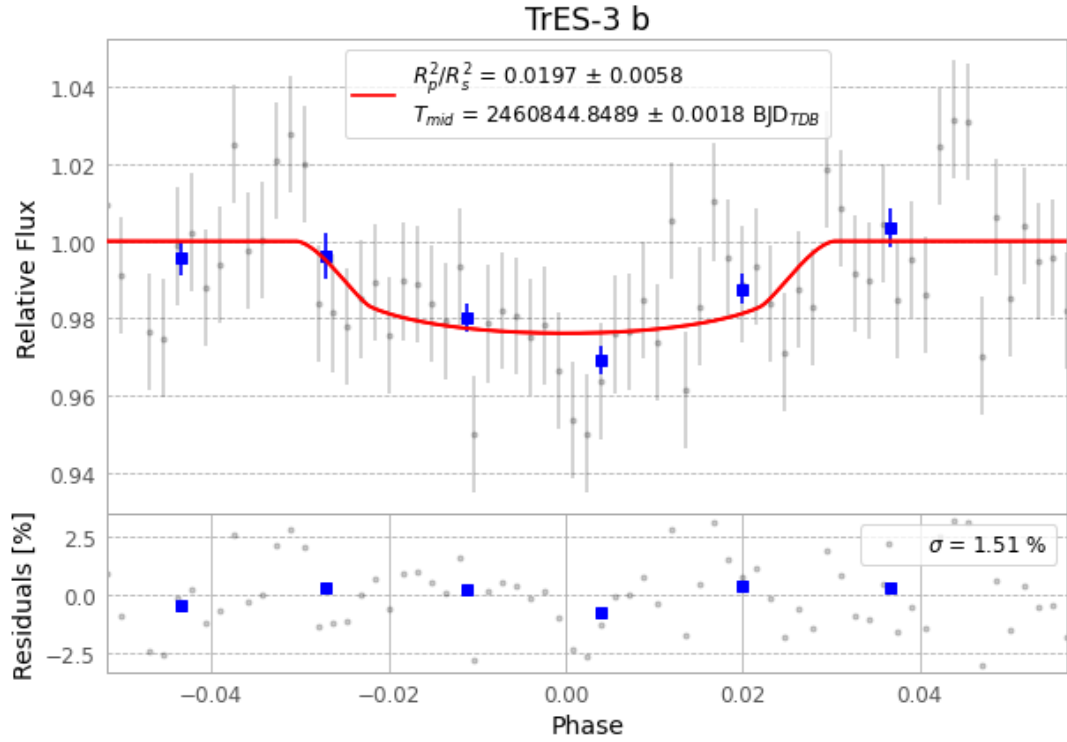


Figure 1 — FinalLightCurve_TrES-3 b_2025-06-17.png (SHA-256 7a90402c3b9de620...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 244], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a5facd199a3f2fb3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

69 FITS frames (45 MB), TRES-3250618063917.FITS → TRES-3250618100317.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250618.log (c446b88da299...), FinalLightCurve_TrES-3 b_2025-06-17.png (7a90402c3b9d...), FinalLightCurve_TrES-3 b_2025-06-17.csv (3c35f5f5c294...)

Compute resources

Wall time 140.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 18:49Z.